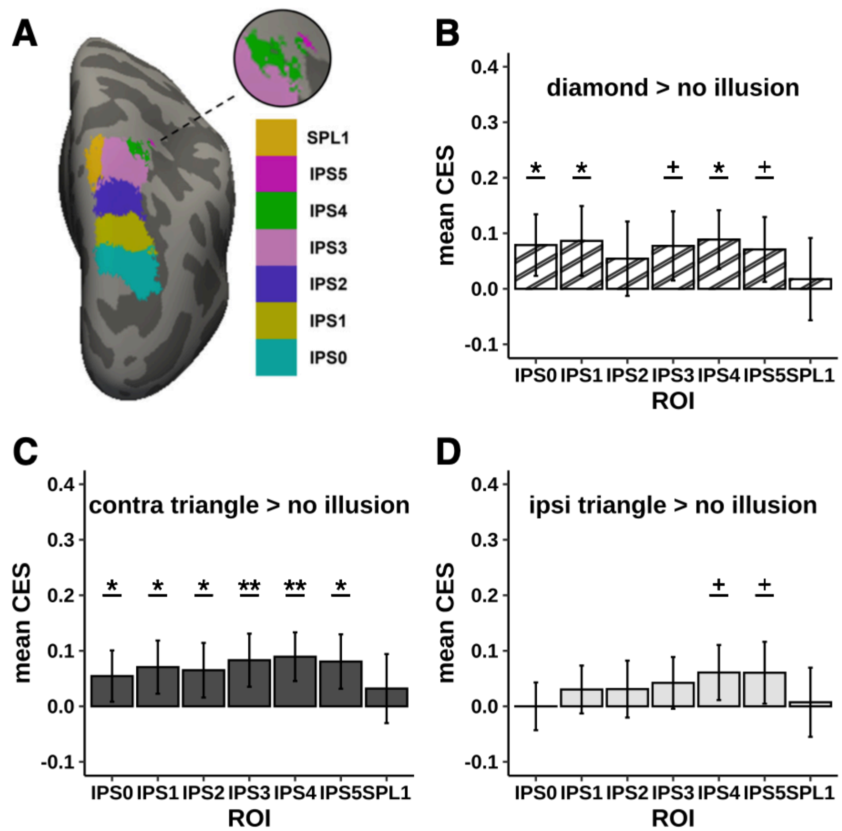
Visualization and reporting

ROI analysis

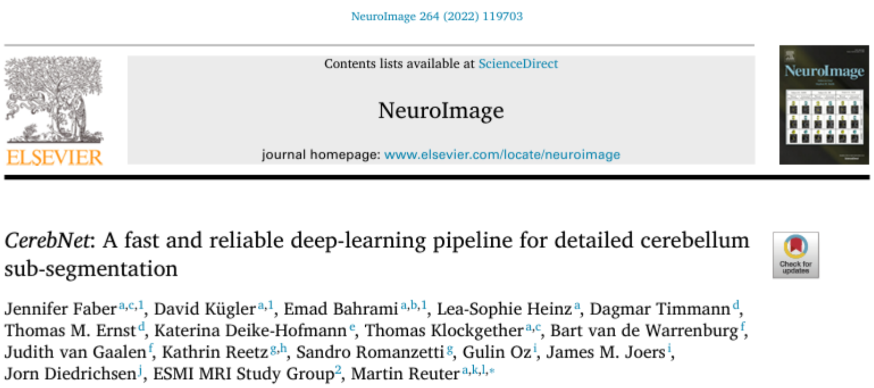


See also: Planning

If you were planning a ROI analysis, you will typically have one value per ROI, subject and condition. You can report descriptive statistics graphically using your favourite approach (violin plots, bar plots, boxplots, etc.).

Usually we also visualize ROIs at their locations.

Example



NeuroImage 264 (2022) 119703

Contents lists available at ScienceDirect

NeuroImage

journal homepage: www.elsevier.com/locate/neuroimage

CerebNet: A fast and reliable deep-learning pipeline for detailed cerebellum sub-segmentation

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Figure 1: (Faber et al. 2022)

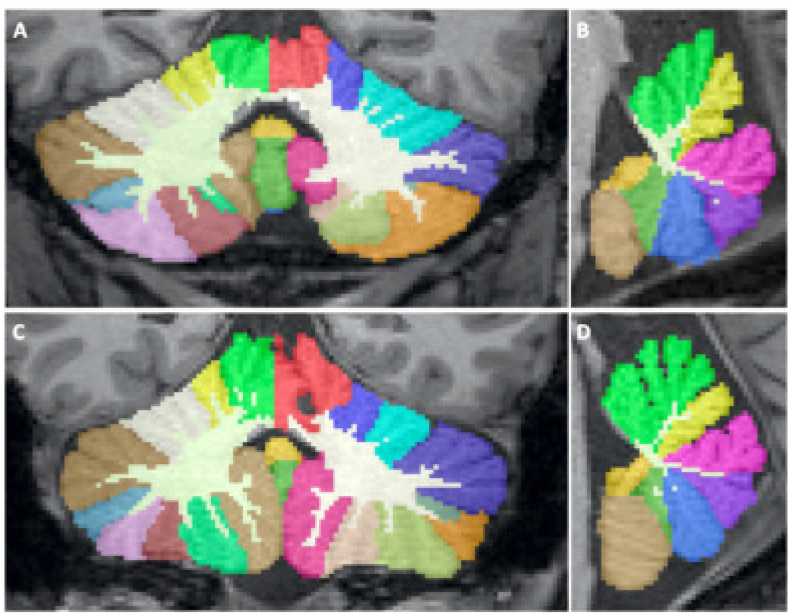


Figure 2: (Faber et al. 2022)

Run the segmentation

Deep-MI/FastSurfer: PyTorch implementation of FastSurferCNN

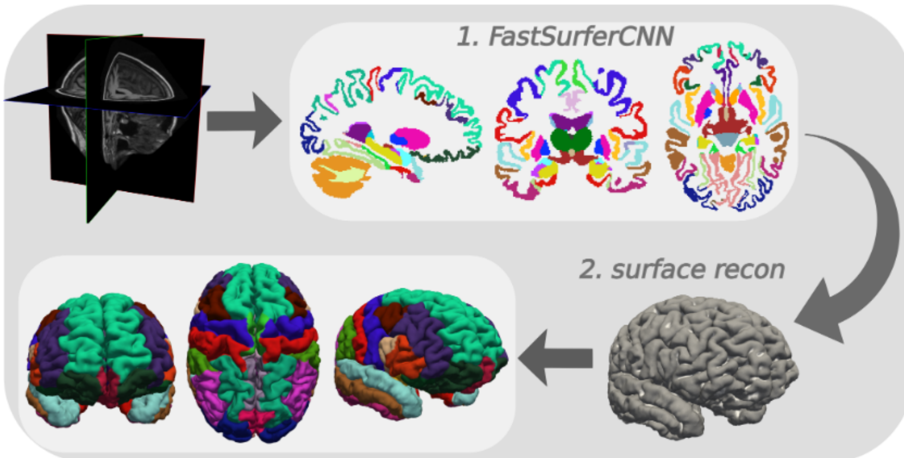


Figure 3: (Henschel et al. 2020)

```
# create the directory
mkdir /home/jovyan/2026_ANI/lobule9_bids_correct/derivatives/fastsurfer

# run the fastsurfer as a singularity container
singularity exec --nv \
                 --no-mount home,cwd -e \
                 -B $HOME/2026_ANI/lobule9_bids_correct:$HOME/my_mri_data \
                 -B $HOME/2026_ANI/lobule9_bids_correct/derivatives/fastsurfer:$HOME/my_fast
                 -B $HOME/license.txt:$HOME/my_fs_license.txt \
                 /shared/singularity/fastsurfer-gpu.sif \
                   /fastsurfer/run_fastsurfer.sh \
                     --fs_license $HOME/my_fs_license.txt \
                     --t1 $HOME/my_mri_data/sub-001/ses-1/anat/sub-001_ses-1_acq-mprage_T1w.
                     --sid sub-001 --sd $HOME/my_fastsurfer_analysis \
                     --3T \
                     --threads 4 \
                     --seg_only
```

View output

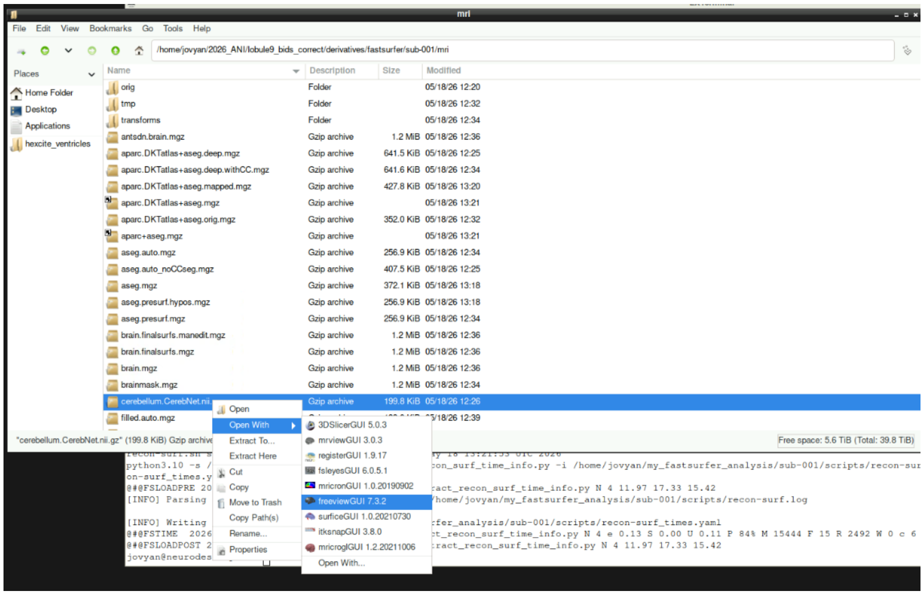


Figure 4: right-click and open the file with freeview

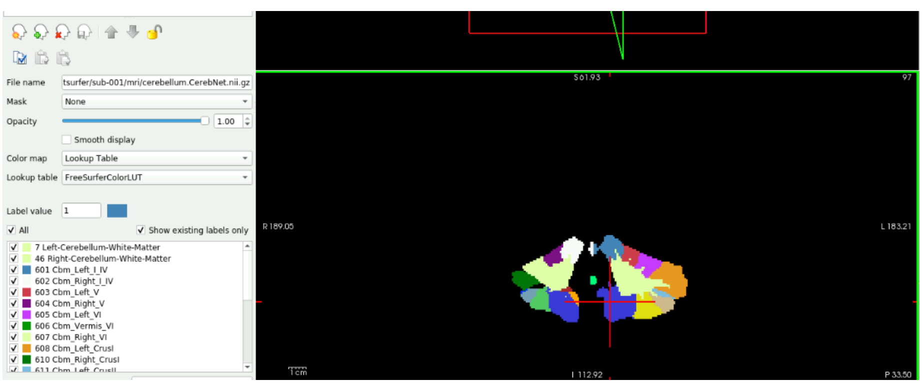
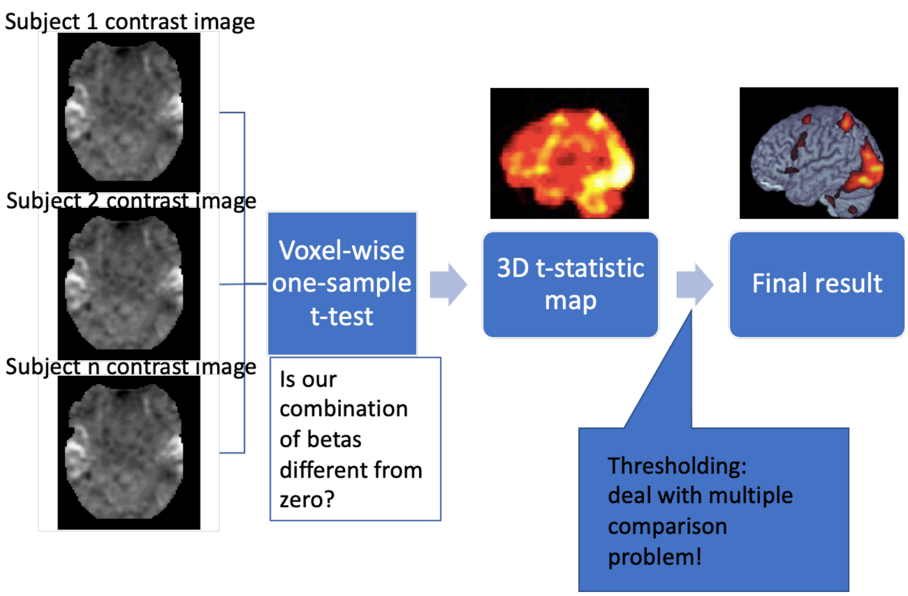


Figure 5: set the color map to "Lookup Table"

Multiple comparison correction

Reminder: group analysis



Family-wise error rate

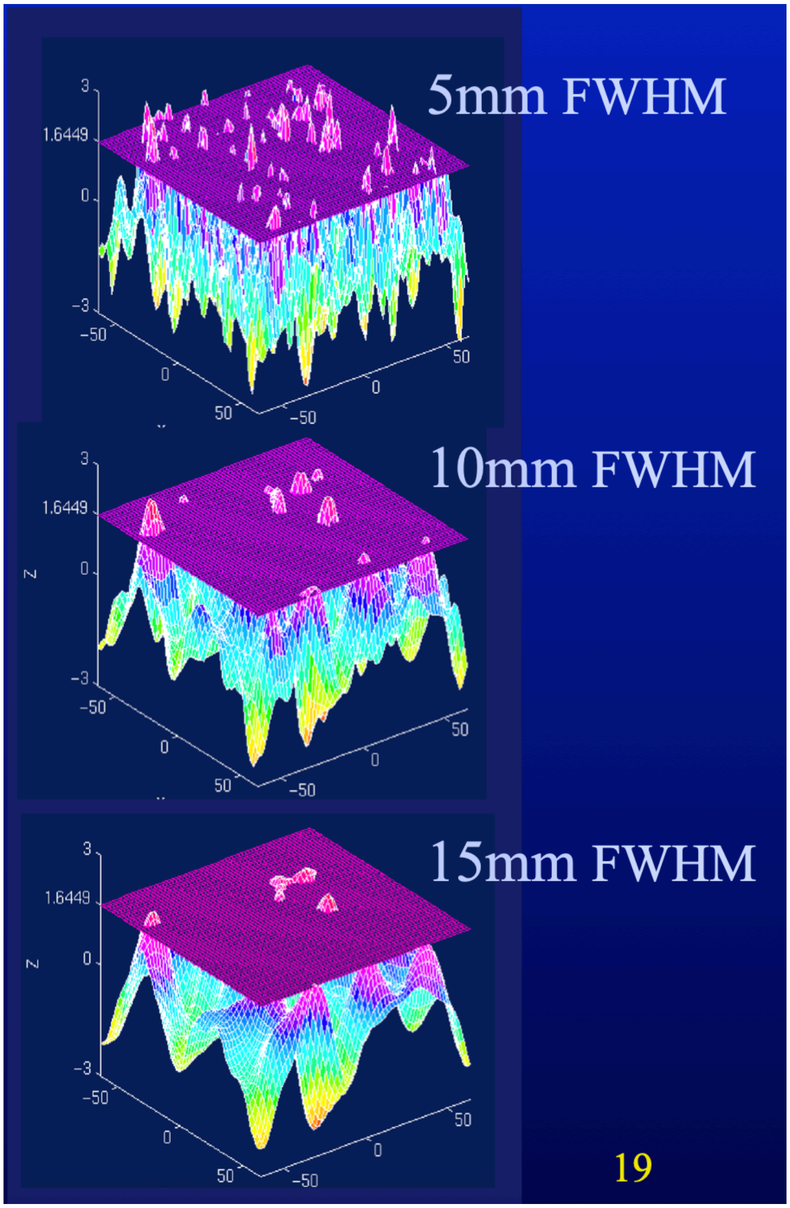


Figure 6: Image credit: unknown

If the data is not smoothed, equivalent to Bonferroni correction

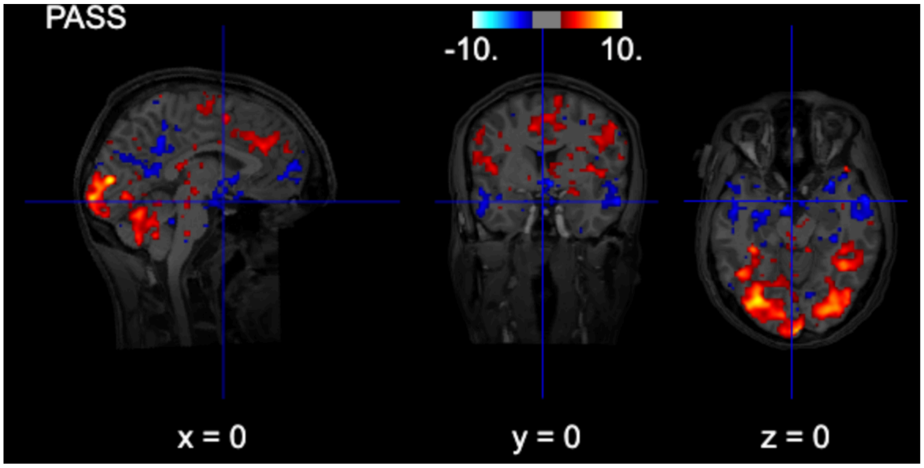
Other correction types

- Cluster threshold

- False discovery rate

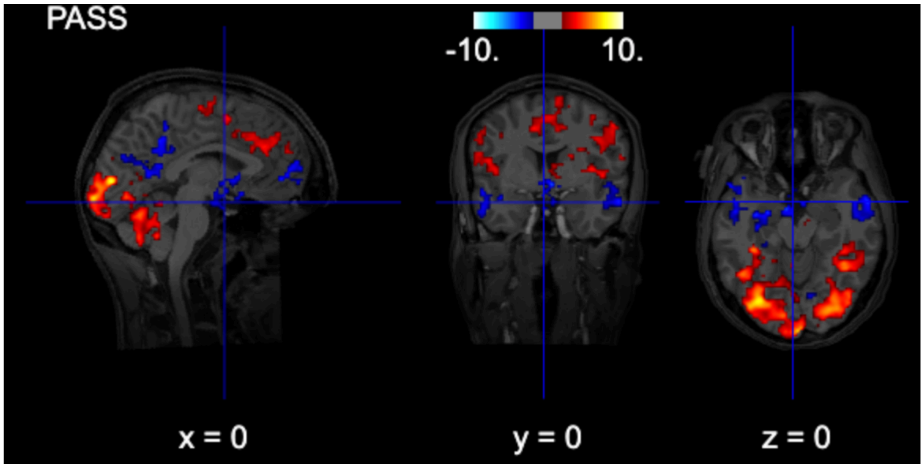
- Cluster-level inference

Uncorrected



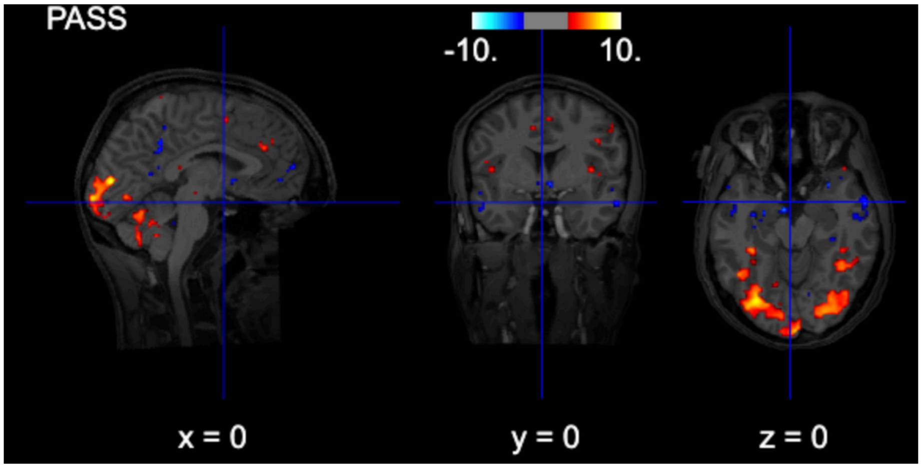

```
# interactive plot (you can browse the activations)
from nilearn import plotting

# Use subject's anatomy as background
bg_img = '/home/jovyan/gambling/bids/derivatives/fmriprep/sub-001/ses-1/anat/sub-001_ses-1_a
plotting.view_img(zmap, threshold=1.96, vmax=10,
    bg_img=bg_img,
    cut_coords=[0, 0, 0],
    width_view=600,
    title=contrast_string)
```

Cluster-thresholded



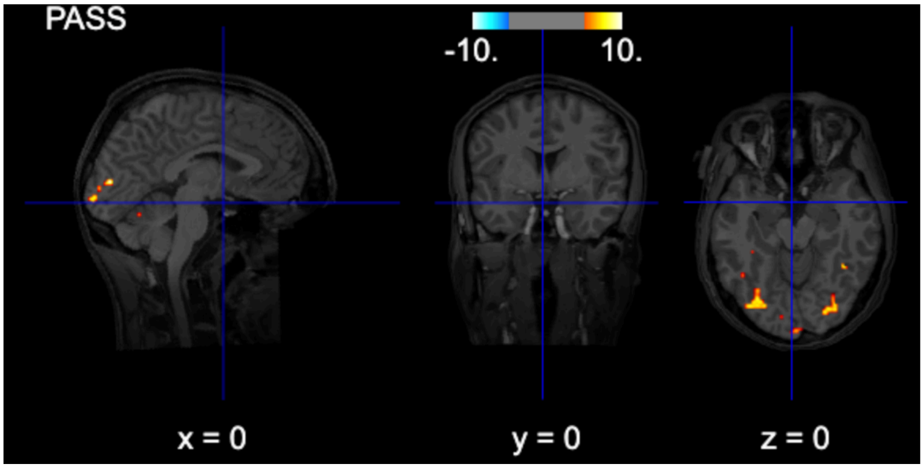

```
from nilearn.glm import threshold_stats_img

# note: fpr with alpha=0.05 is the same as uncorrected z>=1.96 (p<0.05)
thresholded_map1, threshold1 = threshold_stats_img(
    zmap,
    alpha=0.05,
    height_control="fpr",
    cluster_threshold = 100,
    two_sided=True,
)

# Use subject's anatomy as background
bg_img = '/home/jovyan/gambling/bids/derivatives/fmriprep/sub-001/ses-1/anat/sub-001_ses-1_a
plotting.view_img(thresholded_map1, threshold=threshold1, vmax=10,
    bg_img=bg_img,
    cut_coords=[0, 0, 0],
    width_view=600,
    title=contrast_string)
```

FDR-thresholded




```
from nilearn.glm import threshold_stats_img

thresholded_map1, threshold1 = threshold_stats_img(
    zmap,
    alpha=0.05,
    height_control="fdr",
    two_sided=True,
)

# Use subject's anatomy as background
bg_img = '/home/jovyan/gambling/bids/derivatives/fmriprep/sub-001/ses-1/anat/sub-001_ses-1_a
plotting.view_img(thresholded_map1, threshold=threshold1, vmax=10,
    bg_img=bg_img,
    cut_coords=[0, 0, 0],
    width_view=600,
    title=contrast_string)
```

FWE-thresholded

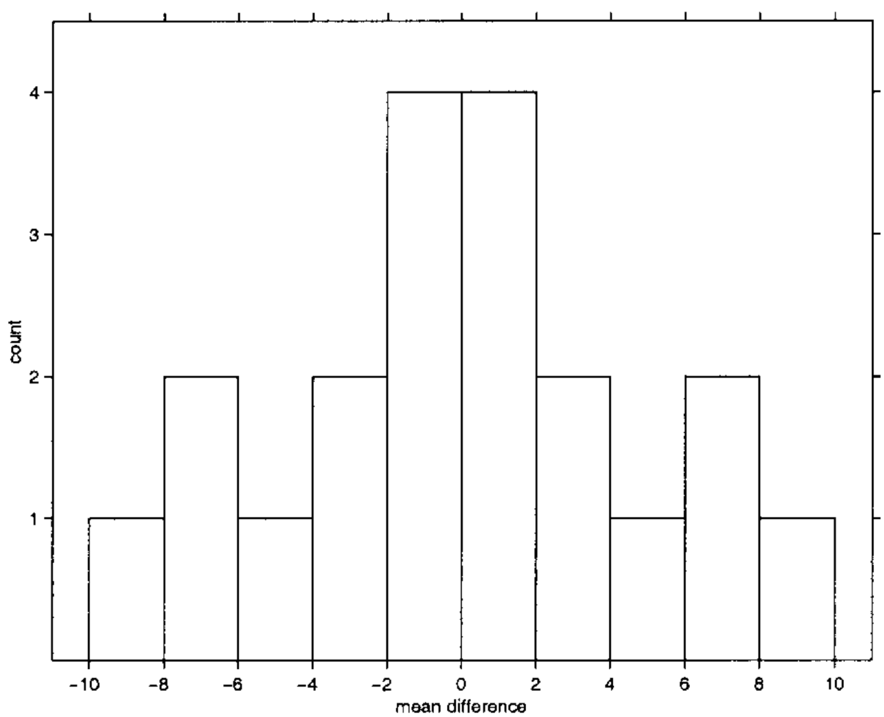



```python
from nilearn.glm import threshold_stats_img

thresholded_map1, threshold1 = threshold_stats_img(
    zmap,
    alpha=0.05,
    height_control="bonferroni",
    two_sided=True,
)

# Use subject's anatomy as background
bg_img = '/home/jovyan/gambling/bids/derivatives/fmriprep/sub-001/ses-1/anat/sub-001_ses-1_a
plotting.view_img(thresholded_map1, threshold=threshold1, vmax=10,
    bg_img=bg_img,
    cut_coords=[0, 0, 0],
    width_view=600,
    title=contrast_string)
```

Non-parametric inference

Permutation-based inference

1. AAABBB
2. AABABB
3. AABBAB
4. AABBBA
5. ABAABB
6. ABABAB
7. ABABBA
8. ABBAAB
9. ABBABA
10. ABBBAA
11. BAAABB
12. BAABAB
13. BAABBA
14. BABAAB
15. BABABA
16. BABBAA
17. BBAAAB
18. BBAABA
19. BBABAA
20. BBBAAA



An example difference between conditions A and B. The labels of each trial are permuted and each time a test statistic (in this example mean difference) is computed. The empirical likelihood to observe a specific value is derived. (Nichols and Holmes 2001)

For the multiple comparison case, we use the same logic, but instead of single voxel we consider the maximum of each statistical map.

Result reporting

References

Faber, Jennifer, David Kügler, Emad Bahrami, Lea-Sophie Heinz, Dagmar Timmann, Thomas M. Ernst, Katerina Deike-Hofmann, et al. 2022. "CerebNet: A Fast and Reliable Deep-Learning Pipeline for Detailed Cerebellum Sub-Segmentation." *NeuroImage* 264 (December): 119703. https://doi.org/10.1016/j.neuroimage.2022.119703.

Henschel, Leonie, Sailesh Conjeti, Santiago Estrada, Kersten Diers, Bruce Fischl, and Martin Reuter. 2020. "FastSurfer - A Fast and Accurate Deep Learning Based Neuroimaging Pipeline." *NeuroImage* 219 (October): 117012. https://doi.org/10.1016/j.neuroimage.2020.117012.

Nichols, Thomas E., and Andrew P. Holmes. 2001. "Nonparametric Permutation Tests for Functional Neuroimaging: A Primer with Examples." *Human Brain Mapping* 15 (1): 1–25. https://doi.org/10.1002/hbm.1058.